Practical Assignment 4: Classification Models

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GitHub Link: https://github.com/sakshishewale01/AI_LAB_PRACTICALS.git

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1.Title

Develop classification models (SVM, Decision Trees, KNN, and Naïve Bayes), evaluate performance using relevant metrics (accuracy, precision, recall, F1-score, loss) and summarize the findings of comparative

2. Introduction and Objectives

Classification assigns an input to one of several predefined classes. Many algorithms exist, and each makes different assumptions about the data, so choosing one requires evidence rather than habit. The objectives of this assignment are:

1. To develop classification models using SVM, Decision Tree, KNN and Naïve Bayes.
2. To evaluate them with accuracy, precision, recall, F1-score and loss.
3. To summarise a comparative analysis and discuss the suitability of each algorithm for different classification tasks.

3. Datasets

All datasets are bundled with scikit-learn, so the experiment is fully reproducible and free to run in Google Colab.

Dataset	Samples	Features	Classes	Role in the study
Breast Cancer Wisconsin	569	30 numeric	2 (212 malignant, 357 benign)	Main binary experiment
Wine	178	13 numeric	3	Small multiclass dataset

Dataset	Samples	Features	Classes	Role in the study
Digits	1,797	64 numeric (8×8 pixel images)	10	High-dimensional multiclass dataset

The Breast Cancer dataset has no missing values. Its classes are moderately imbalanced (about 63 % benign), so stratified splitting was used throughout.

4. Methodology

4.1 Pre-processing and data split

Each dataset was split 80 % training and 20 % testing with stratification (`random_state = 42`). For Breast Cancer this gave 455 training and 114 test samples (42 malignant, 72 benign). Feature scaling with `StandardScaler` was placed inside a scikit-learn `Pipeline`, so the scaler is fitted on training data only and no information leaks from the test set. Scaling is essential for SVM and KNN, which rely on distances or margins, and harmless for DT and NB.

4.2 Models and hyper-parameter tuning

Each model was tuned with `GridSearchCV` using 5-fold stratified cross-validation on the training set, optimising F1-score. The search spaces and the best values found on Breast Cancer are shown below.

Model	Parameters searched	Best parameters (Breast Cancer)	Best CV F1
SVM	kernel {linear, rbf}, C {0.1, 1, 10, 100}, gamma {scale, 0.01, 0.1}	linear kernel, C = 0.1	0.9810
Decision Tree	criterion {gini, entropy}, max_depth {2,3,4,5,7,None}, min_samples_leaf {1,3,5,10}	entropy, max_depth = 4, min_samples_leaf = 5	0.9493
KNN	k {1,3,5,7,9,11,15}, weights {uniform, distance}, metric {euclidean, manhattan}	k = 3, uniform weights, Manhattan distance	0.9774
Gaussian Naïve Bayes	var_smoothing $\{10^{-11} \ldots 10^{-1}\}$	var_smoothing = 0.1	0.9535

4.3 Evaluation metrics

With TP, TN, FP and FN denoting true/false positives and negatives (class 1, benign, is the positive class):

- **Accuracy** = (TP + TN) / (TP + TN + FP + FN): overall share of correct predictions.

- **Precision** = TP / (TP + FP): of the samples predicted positive, how many really are.
- **Recall** = TP / (TP + FN): of the real positives, how many were found.
- **F1-score** = 2 × Precision × Recall / (Precision + Recall): harmonic mean of the two.
- **Loss (log-loss / cross-entropy)** = $-(1/N)\ \Sigma\ \Sigma\ y_{i},c \cdot \log(p_{i},c)$: penalises wrong predictions made with high confidence; lower is better. It measures the quality of the predicted probabilities, not only of the final label.
- **ROC-AUC** was also reported as a threshold-independent measure.

For the multiclass datasets (Wine, Digits), precision, recall and F1 are macro-averaged.

5. Algorithms in Brief

Support Vector Machine. Finds the hyper-plane that maximises the margin between classes; kernels (e.g. RBF) let it learn non-linear boundaries. C controls the trade-off between a wide margin and training errors.

Decision Tree. Repeatedly splits the data on the feature and threshold that most reduce impurity (Gini or entropy), producing human-readable if-then rules. Depth and minimum leaf size control over-fitting.

K-Nearest Neighbours. A lazy learner that stores the training data and classifies a new point by majority vote among its k closest training points. It is sensitive to the choice of k and to feature scaling.

Gaussian Naïve Bayes. Applies Bayes' theorem with the "naïve" assumption that features are conditionally independent given the class, and models each feature with a Gaussian distribution. Training is a single pass over the data.

6. Results on the Main Task (Breast Cancer, Test Set)

Model	Accuracy	Precision	Recall	F1-score	Log-loss	ROC-AUC
SVM	**0.9825**	**0.9861**	**0.9861**	**0.9861**	**0.0858**	**0.9937**
Decision Tree	0.8772	0.9531	0.8472	0.8971	1.0652	0.9593
KNN	0.9649	0.9595	0.9861	0.9726	0.6836	0.9714
Naïve Bayes	0.9474	0.9459	0.9722	0.9589	0.3763	0.9878

6.1 Error analysis (114 test samples)

Model	Correct	Errors	Malignant missed (predicted benign)	Benign wrongly flagged (predicted malignant)
SVM	112	2	1	1
KNN	110	4	3	1

Model	Correct	Errors	Malignant missed (predicted benign)	Benign wrongly flagged (predicted malignant)
Naïve Bayes	108	6	4	2
Decision Tree	100	14	3	11

In a medical setting, missing a malignant case is the costlier error. SVM missed only one malignant tumour out of 42; the Decision Tree caught 39 of 42 malignant cases (recall 0.9286) but at the price of 11 false alarms, which explains its low benign recall (0.8472) and low malignant precision (0.78).

6.2 Robustness: 5-fold cross-validation on the full dataset (mean ± std)

Model	Accuracy	Precision	Recall	F1	Log-loss
SVM	0.9754 ± 0.0170	0.9683 ± 0.0290	0.9944 ± 0.0069	0.9809 ± 0.0130	0.0886 ± 0.0463
KNN	0.9666 ± 0.0188	0.9652 ± 0.0287	0.9833 ± 0.0162	0.9738 ± 0.0146	0.4300 ± 0.3719
Naïve Bayes	0.9333 ± 0.0196	0.9284 ± 0.0224	0.9693 ± 0.0283	0.9479 ± 0.0155	0.6896 ± 0.3832
Decision Tree	0.9227 ± 0.0349	0.9214 ± 0.0553	0.9636 ± 0.0272	0.9405 ± 0.0248	1.1636 ± 0.2466

The ranking is identical to the single-split result (SVM > KNN > NB > DT), so it is not a lucky split. The Decision Tree also has the largest standard deviation in accuracy (±0.035), showing its instability.

6.3 Analysis of loss and bias–variance behaviour

- **Decision Tree:** training log-loss falls to nearly zero as depth increases while test log-loss rises sharply, from about 0.3 at depth 1 to above 3 at depth 6 and beyond. This is classic over-fitting: deep trees memorise the training data and output probabilities of exactly 0 or 1, so every misclassification is punished very heavily.
- **KNN:** k = 1 gives zero training error but a high and unstable test error; test error is lowest for small-to-moderate k (about 3 to 9) and drifts upward again for large k as the model under-fits. Small k also produces coarse probabilities (multiples of 1/k), hence the relatively high log-loss.
- **SVM:** for very small C the model under-fits (error about 37 %); error drops sharply by C ≈ 0.03 and is lowest around C ≈ 1, then the gap between training and test error widens slightly for very large C.
- **Naïve Bayes:** has few parameters and therefore low variance, but the independence assumption is violated by the strongly correlated cell-measurement features (radius,

perimeter and area are almost the same information). This makes its probabilities over-confident and its loss (0.376) higher than the SVM's.

7. Generalisation to Other Classification Tasks

The same pipeline (re-tuned per dataset, macro-averaged metrics) was applied to Wine and Digits.

Dataset	Model	Accuracy	Precision	Recall	F1-score	Log-loss
Wine	SVM	0.9722	0.9778	0.9667	0.9710	0.0924
Wine	Decision Tree	0.9444	0.9583	0.9389	0.9457	2.0024
Wine	KNN	1.0000	1.0000	1.0000	1.0000	0.0937
Wine	Naïve Bayes	0.9722	0.9744	0.9762	0.9743	0.0293
Digits	SVM	0.9750	0.9760	0.9748	0.9748	0.1181
Digits	Decision Tree	0.8028	0.8039	0.8019	0.8014	5.0151
Digits	KNN	0.9667	0.9678	0.9664	0.9663	0.6337
Digits	Naïve Bayes	0.8917	0.8947	0.8908	0.8918	1.2879

Average F1 rank across the three tasks (1 = best): SVM 1.67, KNN 1.67, Naïve Bayes 2.67, Decision Tree 4.00.

Key observations:

- On the very small Wine set (178 samples) every model scores above 94 %, and differences are within the noise of a 36-sample test set. KNN reached 100 % and Naïve Bayes achieved the lowest loss (0.029), showing that simple models are perfectly adequate when classes are well separated.
- On Digits (64 features, 10 classes) the gap widens: SVM (97.5 %) and KNN (96.7 %) clearly beat Naïve Bayes (89.2 %) and the Decision Tree (80.3 %). Neighbouring pixels are highly correlated, which hurts Naïve Bayes, and axis-aligned splits of single pixels are a poor way to describe shapes, which hurts the tree.
- The Decision Tree's loss is extremely high on Wine (2.00) and Digits (5.02) because of its hard 0/1 leaf probabilities, so it should not be used where well-calibrated probabilities are needed unless it is ensembled or calibrated.

8. Comparative Analysis and Suitability

Criterion	SVM	Decision Tree	KNN	Naïve Bayes
Accuracy in this	Highest or near-highest on all	Lowest on	Very high, best	Good on small / low-dimensional

Criterion	SVM	Decision Tree	KNN	Naïve Bayes
study	tasks	all tasks	on Wine	data, weaker on Digits
Loss / probability quality	Best on the two larger tasks	Worst everywhere	Moderate to poor (coarse probabilities)	Best on Wine, over-confident on correlated features
Training speed	Slowest of the four on Digits (0.18 s) but still fast here	Fast	Almost none (lazy learner)	Fastest
Prediction speed	Fast	Very fast	Slowest as data grows	Very fast
Needs feature scaling	Yes	No	Yes	No
Interpretability	Low (high for linear kernel weights)	**Highest** (explicit rules)	Medium (nearest examples)	Medium (class-conditional statistics)
Sensitive to noise / outliers	Moderate	High (unstable)	High	Low
Handles high dimensions	**Very well**	Moderately	Poorly (curse of dimensionality)	Well, if features are independent
Works with very small data	Well	Poorly	Well	**Very well**

8.1 When to choose each algorithm

SVM is the best general-purpose choice for medium-sized, numeric, high-dimensional problems where accuracy matters, such as medical diagnosis from measurements, image classification and text with TF-IDF features. Its drawbacks are the need for scaling and tuning, training time that grows quickly beyond tens of thousands of samples, and low interpretability.

Decision Tree is suitable when the decision must be explained to a human, for example loan approval rules, clinical triage protocols or regulatory settings, and when data contains mixed numeric and categorical features. On its own it over-fits and is unstable, so for accuracy it is normally used as the base learner of Random Forests or Gradient Boosting.

KNN is a strong, assumption-free baseline for small and medium, low-dimensional, well-clustered data (recommendation, pattern recognition, anomaly detection), and it adapts instantly

when new data is added. It becomes slow and less accurate when the dataset is large, high-dimensional or contains irrelevant features.

Naïve Bayes is the right tool for text classification (spam filtering, sentiment analysis), real-time prediction, very small training sets and extremely high-dimensional sparse data, where its speed and low variance outweigh the independence assumption. It is a poor choice when features are strongly correlated and calibrated probabilities are required.

9. Conclusion

No single algorithm is best for every problem, but the experiments give a clear, evidence-based ordering for numeric tabular data. SVM delivered the best balance of accuracy, precision, recall, F1-score and loss on the main task (98.25 % accuracy, F1 0.986, log-loss 0.086) and generalised well to the multiclass tasks. KNN was nearly as accurate but its probabilities were coarser and it becomes costly at prediction time on large data. Naïve Bayes gave a good accuracy-to-cost ratio and excelled on the smallest dataset, but suffered when features were correlated. The Decision Tree gave the lowest accuracy and highest loss because of over-fitting, yet remains the only model whose reasoning can be read directly, which is decisive in some applications.

The practical recommendation is to start with a scaled SVM or KNN as a strong baseline, use Naïve Bayes when speed, tiny data or text is the priority, use a Decision Tree when interpretability is a requirement, and always compare models using several metrics (including loss) with cross-validation rather than accuracy alone.

10. Limitations and Future Work

- Results come from three small, clean, built-in datasets; performance on noisy, imbalanced or very large real-world data may differ.
- Only Gaussian Naïve Bayes was used; Multinomial or Bernoulli variants suit text and count data better.
- Future work: calibrate probabilities (Platt or isotonic), try ensemble methods (Random Forest, XGBoost), handle class imbalance with SMOTE or class weights, and test on larger text datasets.

11. Tools and References

- Python 3, scikit-learn 1.8, pandas, NumPy, matplotlib, seaborn (Google Colab, free tier).
- Companion file: `Classification_Comparative_Analysis.ipynb` (complete code, plots, confusion matrices, ROC curves, learning curves and the decision-tree diagram).

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